



Faculty of Industrial Engineering

INGB 413: GA Exam Individual Assignment

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Application of ECSA's Graduate Attribute 4

ECSA Graduate Attribute 4: Demonstrate competence to conduct investigations of complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

The Table below outlines exactly how GA 4 was applied in this assignment.

GA Component	Application
Research method.	While the report does not explicitly state the research method used, it is evident that the Six Sigma DMAIC (Define, Measure, Analyse, Improve, Control) methodology was applied, with slight adjustments made to the Improve and Control phases. Although an improvement is clearly proposed, the focus was more on recommending a better option between the two processes, and the Control phase has yet to be implemented.
Research-based knowledge.	Throughout the report, academic journals, textbooks, and industry sources were used to deepen the understanding of quality control methods and statistical process control. This research also guided the selection of appropriate tools and techniques for analysis.
Design of experiments.	Guidelines were provided for conducting the two experimental processes, but the specific experimental procedures were designed based on individual prior research and personal interpretation.
Analysis and interpretation of data.	Data collected independently from two different experiment processes were analysed through control tools such as control charts, pareto charts and process capability (DPMO- defects per million opportunities), and interpreted with researched guidelines.
Synthesis of information to provide valid conclusions.	Interpretations of the two different processes were compared to draw meaningful conclusions and select the better process.

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1. Background

Tangram Inc. is a company that focuses on producing tangram puzzles. A tangram puzzle is a traditional Chinese puzzle made up of seven flat geometric shapes. Tangra Inc. sources each puzzle piece from a different supplier. After receiving the puzzle pieces, they are assembled by hand into a wooden board and then these completed tangram sets are shipped to customers. Figure 1 below illustrates a completed tangram puzzle.

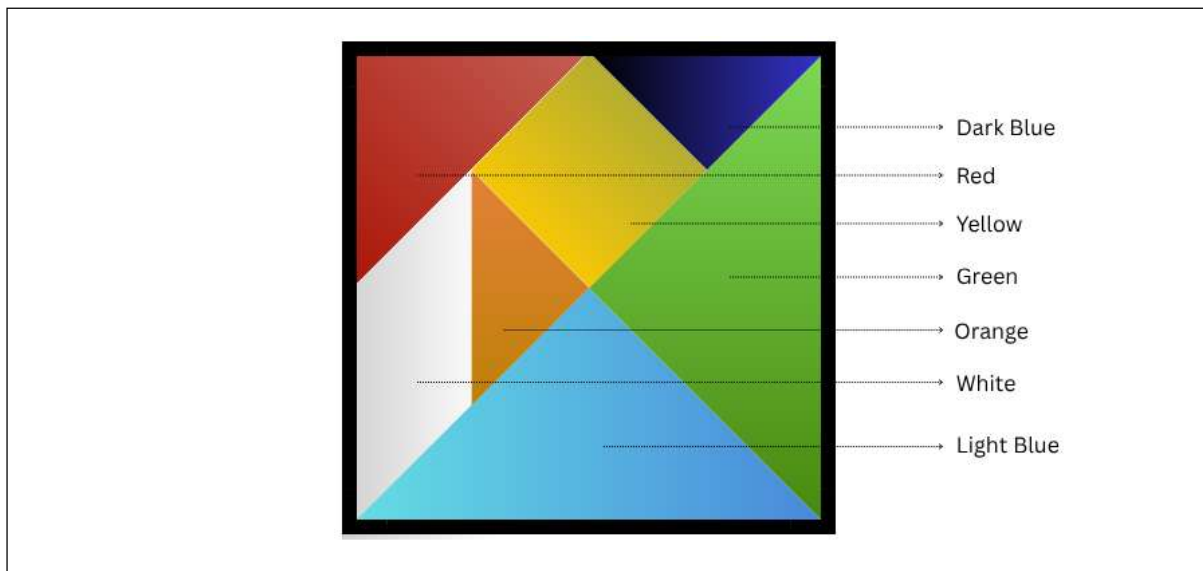


Figure 1: Traditional Chinese Tangram Puzzle

Figure 1 clearly illustrates that each puzzle piece is required to be the correct shape and size to fit into the wooden board, otherwise it results in an incomplete tangram set. In *Introduction to Statistical Quality Control* (Montgomery, 2013), a clear distinction is made between a *defect* and a *defective*. A *defect*, or nonconformity, refers to an instance where a product fails to meet a specified requirement, but this does not necessarily make the product unusable. In contrast, a *defective* product is one where the defect(s) are serious enough to compromise its safe or effective use. In other words, while all defectives contain defects, not all defects render a product defective.

Therefore, we can conclude that if a puzzle piece does not meet the required shape or size, it is classified as a defect because it fails to meet the specified criteria. However, this does not necessarily mean the piece is unusable in other contexts. In the case of a tangram set, if even one defective piece is included on the wooden board, the final product, a complete tangram set, would be considered defective, as it hinders proper use.

2. Problem Statement

Tangram Inc. has struggled to meet demand due to a rise in nonconforming puzzle pieces, resulting in more defective tangram sets.

3. Aim and Objectives

3.1 Aim

The aim of this investigation is to analyse two assembly processes and select the process that delivers a higher quality outcome while considering cost implications.

3.2 Objectives

To achieve the aim the following objectives are defined:

1. To define two assembly processes in detail.
2. To collect data from both assembly processes.
3. To analyse the data using quality management and statistical process control techniques.
4. To identify the better assembly process for detecting defective puzzle pieces, balancing quality and cost.

4. Quality Requirement

The quality management team at Tangram Inc. has specified a minimum performance standard for the assembly process, requiring it to operate at a 2-sigma level in terms of defective parts per million.

5. Data Collection

This section provides details on the processes followed and the type of data collected.

5.1 Process Overview

Tangram Inc.'s Quality Management team identified two specific assembly processes for further investigation. These processes were then refined based on insights from relevant literature before being implemented in the experiment. This section provides more detail on both assembly processes.

5.1.1 Assembly Process 1

Sampling raw materials, before production allows for the early identification of defects, which helps prevent defective products later on and reduces costs (Al-Tahat, Alqeed, & Al-Fawares, 2020). "Sampling is defined as the process of selecting a representative group from the

population under study" (Etikan, Musa, & Alkassim, 2016). Thus, in assembly process 1, the operator will sample and inspect the puzzle pieces for defects as an attempt to reduce defective tangram sets at the end of assembly.

Stratified random sampling is a method where the population is divided into distinct subgroups based on specific characteristics, such as colour. Samples are then randomly drawn from each group to ensure representation across all groups (Lohr, 2009). Partial stratified random sampling on the other hand involves selecting samples from only some of these groups, rather than from all of the groups (Thompson, 2012). In this assembly process, although the seven different coloured puzzle pieces are mixed together in one container without physical separation into distinct colour groups, the sampling process requires each sample to include five different colours. This approach mimics partial stratified random sampling by visually dividing the container into colour groups and ensuring that five different colour groups were represented in each sample.

5.1.1.1) Assembly Process 1 Steps

The steps followed by the operator is outlined below:

- 1. Raw Materials Receiving:**

The operator receives raw material in bulk. All seven tangram puzzle pieces are mixed together in a single bulk container, without any sorting.

- 2. Sampling:**

The operator selects a sample of five puzzle pieces, each of a different colour, using *partial stratified random sampling* from the container.

- 3. Defect Identification Process 1 and Removal:**

The five sampled puzzle pieces are inspected for defects using a visual comparison method. The operator places each piece onto a pre-sized template board to assess fit. Any piece that does not fit the template precisely is classified as a defect and is removed from the sample.

- 4. Assembly:**

After defect removal, the operator proceeds to assemble the tangram puzzle using the remaining pieces from the original material pool and newly selected puzzle pieces from the container. No further inspection is performed on the newly selected puzzle pieces during the assembly.

5.1.1.2) Assembly Process 1 Metrics:

The metrics collected from assembly process 1 will be:

- Sample Number
- Sample Size (consistent $n=5$)
- Number of Defects in Sample
- Specific Block Defect(s)
- Final Product Defective (Yes/No)
- Number of Defects in Final Puzzle
- Specific Defects in Final Puzzle

5.1.2 Assembly Process 2

Assembly process 2 does not involve any inspection before assembly. The different blocks are however divided into subgroups based on colour, and the operator will select each puzzle piece randomly.

5.1.2.1) Process 2 Steps

The steps followed by the operator is outlined below:

1. Raw Materials:

The operator receives raw material already sorted by colour in lin bins at the assembly station.

2. Picking and Placing:

The operator picks puzzle pieces from the lin bins and places it on the wooden board one at a time.

3. Defect Identification Process 2 and Discarding:

If the operator places a defect, identified through visual detection, the entire assembly (wooden board with puzzle pieces) is discarded, and the tangram set is seen as defective.

5.1.2.2) Process 2 Metrics:

The metrics collected from process 2 was:

- Sample Number
- Sample Size (Pieces Placed Including Defect)
- Number of Defects (1/0)
- Specific Block Defect

5.2 Process Comparison Summary

Figure 2 is a high-level visual comparison between assembly process 1 and 2, to indicate the core difference between the two. Assembly process 1 identifies raw material defects before assembly

and only discards the puzzle pieces that are defects, while assembly process 2 identifies raw material defects while assembling, and discards the entire defective semi-assembled tangram set.

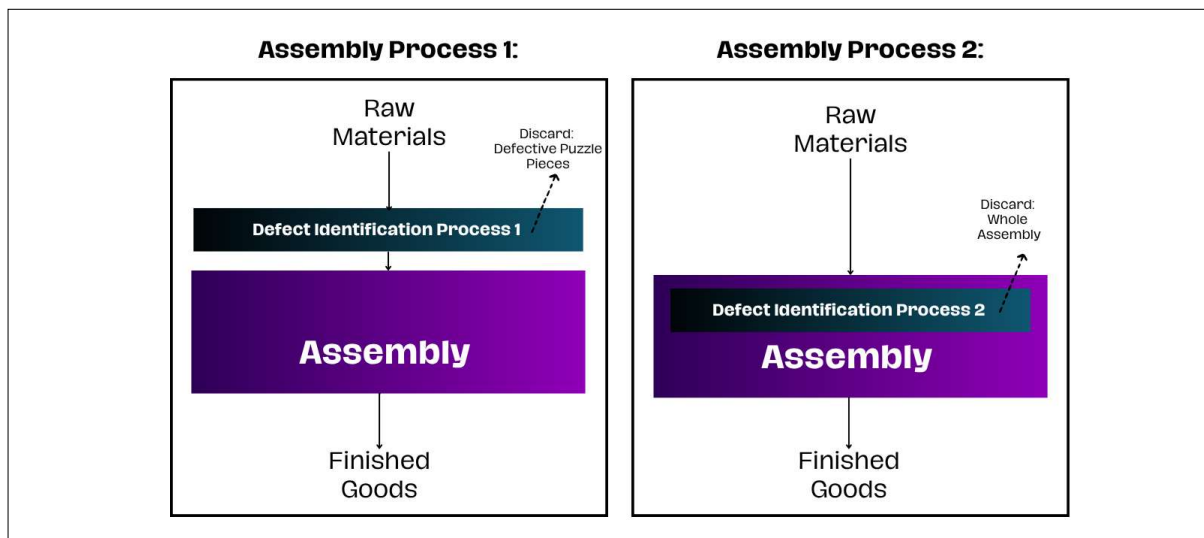


Figure 2: Process Comparison

5.3 Findings

This section outlines the findings from the experiments for both assembly processes.

5.3.1 Assembly process 1

Table 1 provides data collected from assembly process 1 during the 15-minute time limit.

Table 1: Process 1 Collected Data

Sample number	Sample size	Number of defects in sample	Specific Block Defect	Final product defective	How many defects in final puzzle	Specific defects in final puzzle		
1	5	1	Dark blue	YES	2	Dark blue	Yellow	-
2	5	1	Orange	YES	2	Orange	Dark blue	-
3	5	2	White	YES	1	Dark blue	-	-
			Orange					
4	5	3	Dark blue	YES	1	Dark blue	-	-
			Red					
			Orange					
5	5	0	-	NO	0	-	-	-
6	5	1	Dark blue	NO	0	-	-	-
7	5	2	Orange	YES	1	White	-	-
			White					
8	5	2	White	YES	3	Yellow	White	Orange
			Dark blue					
9	5	1	Yellow	YES	2	Dark blue	Orange	-
10	5	0	-	YES	1	Dark blue	-	-
11	5	1	Orange	NO	0	-	-	-
12	5	2	Red	YES	2	Dark blue	White	-
			Orange					
13	5	0	-	YES	1	Orange	-	-
14	5	0	-	YES	1	Orange	-	-
15	5	2	Orange	YES	2	Dark blue	Orange	-
			Yellow					
Total	75	18	-	-	19	-	-	-

5.3.2 Assembly process 2

Table 2 provides data collected from assembly process 2 during the 15-minute time limit.

Table 2: Process 2 Collected Data

Sample number	Sample size (pieces placed including defect)	Number of defects (1/0)	Specific block defect
1	2	1	Dark blue
2	7	0	-
3	3	1	Orange
4	3	1	Orange
5	4	1	Yellow
6	4	1	Yellow
7	1	1	Dark blue
8	2	1	White
9	3	1	Orange
10	1	1	Dark blue
11	1	1	Dark blue
12	2	1	White
13	4	1	Yellow
14	2	1	White
15	3	1	Orange
16	1	1	Dark blue
17	1	1	Dark blue
18	4	1	Yellow
19	2	1	White
20	1	1	Dark blue
21	2	1	White
22	3	1	Red
23	3	1	Red
24	4	1	Yellow
25	3	1	Red
26	3	1	Red
27	7	0	-
28	6	1	White
29	3	1	Red
30	6	1	White
31	3	1	Red
32	3	1	Red
33	4	1	Yellow
34	3	1	Red
35	3	1	Red
36	3	1	Red
37	3	1	Red
38	3	1	Red
39	7	1	Dark blue
40	3	1	Red
Total	126	38	-

6. Data Analysis

Tangram Inc. is currently unable to meet demand due to a high number of defective tangram sets. Defects increase product variability, which negatively impacts manufacturing quality, leading to inconsistent products and reduced customer satisfaction (Sadikin, 2023). Quality management can be used to address these issues by monitoring processes and identifying sources of defectives to improve overall quality. The Seven Basic Tools for quality management - check sheet, histogram, Pareto chart, cause-and-effect diagram, scatter diagram, control chart, and flowchart - are useful for identifying and solving quality issues (Neyestani, 2017). Another important tool, though not part of the seven, is process capability, which helps assess how well a process meets specifications and is discussed in Montgomery's *Introduction to Statistical Quality Control* (2019).

Control charts, originally developed by Walter A. Shewhart, are important tools in statistical process control (SPC) used to monitor the stability and performance of processes over time. They consist of a central line representing the process average, along with upper and lower control limits typically set at ± 3 standard deviations from the mean. These limits help distinguish between common cause variation (inherent to the process) and special cause variation (due to external factors), enabling timely corrective actions (Montgomery, 2019).

For this investigation, control charts will be used to analyse the manufacturing process of Tangram Inc. Since the presence of even a single defect renders a tangram set as defective, maintaining strict process control is important. Therefore, instead of the typical ± 3 standard deviations, tighter control limits set at ± 2 standard deviations from the mean will be used. This adjustment increases the sensitivity of the control chart.

To help identify whether or not the control chart is in statistical control we will make use of the Western Electric Rules. The Western Electric Rules are a set of decision rules developed by the Western Electric Company in the 1950s to help identify non-random patterns in control charts, indicating that a process may be out of statistical control due to special cause variation (Western Electric Company, 1956). These rules are widely used in statistical quality control to detect shifts or trends in processes that might not breach control limits but still suggest instability (Montgomery, 2019).

The four primary Western Electric Rules are:

- **Rule 1:** One point plots outside the 3σ control limits.
- **Rule 2:** Two out of three consecutive points fall beyond the 2σ limit on the same side of the mean.
- **Rule 3:** Four out of five consecutive points fall beyond the 1σ limit on the same side of the mean.
- **Rule 4:** Eight consecutive points fall on one side of the centreline.

As stated earlier, the control charts used in this investigation will have 2σ control limits; therefore, the Western Electric Rules will be adjusted accordingly. For example, Rule 1 will now signal an out-of-control condition if a single point falls outside the $\pm 2\sigma$ limits instead of the traditional $\pm 3\sigma$. Similarly, Rules 2 and 3, which rely on detecting points in zones between 1σ and 3σ , may be recalibrated to maintain sensitivity while minimising false alarms (Montgomery, 2019; Western Electric Company, 1956). Despite these adjustments, Rule 4, which flags eight consecutive points on one side of the centreline, remains unchanged as it does not rely on σ zones. The adjusted rules that will be used to analyse control charts are listed below.

- **Rule 1:** One point plots outside the 2σ control limits.
- **Rule 2:** Two out of three consecutive points fall beyond the 1.33σ limit on the same side of the mean.
- **Rule 3:** Four out of five consecutive points fall beyond the 0.67σ limit on the same side of the mean.
- **Rule 4:** Eight consecutive points fall on one side of the centreline.

Pareto charts are quality management tools based on the 80/20 principle, showing that most problems are often caused by a few key factors (Juran, 1988). They display causes in descending order of frequency, making it easier to identify which issues to address first for the greatest impact (Brassard & Ritter, 2010). In manufacturing, Pareto charts help prioritise defect types so that efforts can focus on the most common and costly errors (Oakland, 2014).

For this investigation Pareto charts will be useful in identifying what defects occur the most and therefore causes tangram sets to be defective.

Process capability measures how well a process produces products within specified limits by comparing process variability to tolerance levels, using indices like C_p and C_{pk} (Montgomery, 2019). It helps determine if a process is stable and capable of meeting quality standards or if improvements are needed (Juran & Godfrey, 1999). High process capability indicates consistent, defect-free output (Grant, 2014).

A common method to calculate process capability in Six Sigma is using Defects Per Million Opportunities (DPMO), which measures defects per million chances and standardises process performance (Antony, 2004). DPMO helps convert defect rates into sigma levels, showing how well a process meets specifications (Pyzdek & Keller, 2018). This allows organisations to evaluate process effectiveness and compare against targets like the Six Sigma goal of 3.4 defects per million (Harry & Schroeder, 2006).

Pyzdek & Keller (2018) in *The Six Sigma Handbook* explain that DPMO is a universal metric for defect measurement and is independent of the process distribution, focusing instead on the count of defects. Antony (2004) also discusses DPMO in a way that emphasises its use as a general performance metric for any process, without restricting it to normal or any specific distributions. Some sources mention that while process capability indices (like Cp, Cpk) assume normality, DPMO can be used with any process distribution since it's based on defect counts, not parameter estimates from a distribution (Montgomery, 2019).

For this investigation Defects Per Million Opportunities (DPMO) will thus be used to determine the process capabilities regardless of the collected data's distribution, as the quality management team specified that they require the process (in terms of defective parts per million) to run at a 2-sigma level.

At a 2-sigma level, processes typically produce about 308 537 defects per million opportunities, while a 3-sigma process produce approximately 66 807 defects per million. These values decrease dramatically at higher sigma levels, reaching as low as 3.4 defects per million at a 6-sigma level, reflecting world-class process quality (Harry & Schroeder, 2006; Montgomery, 2019). We can thus conclude that the lower the DPMO is, the more capable a process is. The table below lists the sigma-level along with their corresponding defect count.

Table 3: Sigma-Level & DPMO

Sigma-Level	Defect Count (DPMO)
6-sigma	3.4
5-sigma	233
4-sigma	6210
3-sigma	66807
2-sigma	308538
1-sigma	691462

Data analysis will be divided in two sections: Defect Identification Process Analysis and Assembly Process Analysis.

6.1 Defect Identification Process Analysis

This section of analysis will purely be on the defect identification process within each assembly process. Figure 3 below visually shows which processes within the assembly process will be analysed now and that the focus will logically be on the raw materials, as that is where defects are present.

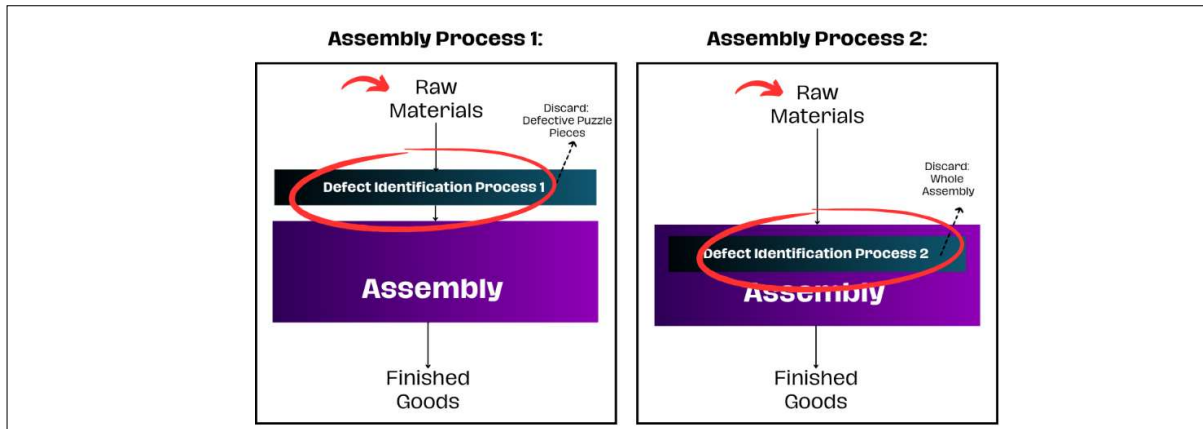


Figure 3: Visual Defect Identification Analysis

6.1.1 Defect Identification Process 1 Analysis

Defects were identified by sampling five puzzle pieces at a time and discarding those that did not meet the required shape or size specifications. The number of defect puzzle pieces in each sample was recorded, as shown in Table 1. An np-chart is an attribute control chart used to track the number of nonconforming units in a process when sample sizes are fixed and the data follows a binomial distribution (Montgomery, 2019). Since each sample consistently contained five pieces, and each piece is either a defect or not, the conditions for using an np-chart are met. Therefore, an np-chart was constructed using the data from Table 1 to monitor the number of defective pieces per sample. For this analysis, the control limits were set at a 2-sigma level, allowing for more sensitive detection of shifts in the quality of the raw puzzle pieces.

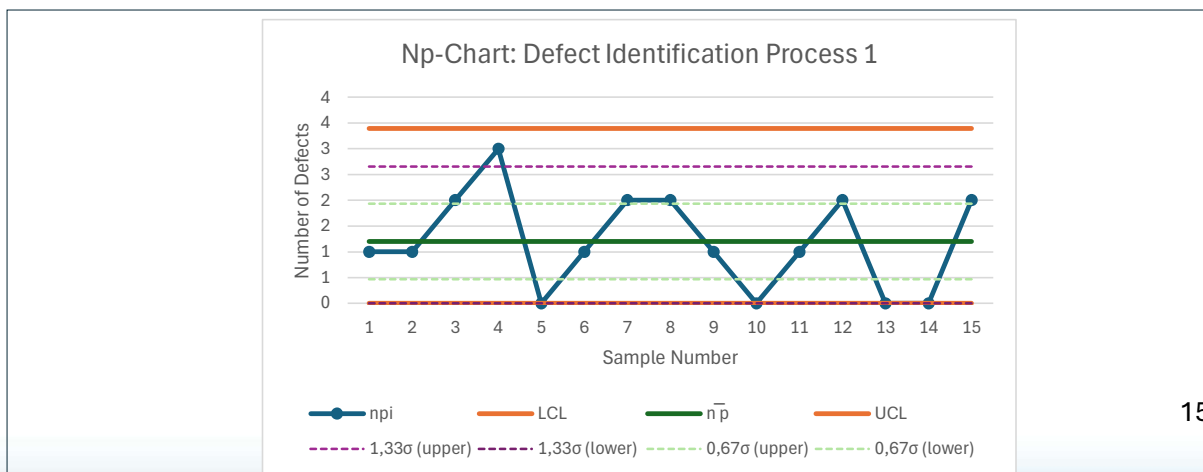


Figure 4: Np-Chart: Defect Identification Process 1

The np-chart for Defect Identification Process 1 displays a lower control limit (LCL) of 0, a central line (CL) of 1.2, and an upper control limit (UCL) of approximately 3.39. Interpretation of the chart, guided by the standard control chart rules, is outlined below:

- Rule 1: No points lie outside the 2σ control limits.
- Rule 2: No two out of three consecutive points fall beyond the 1.33σ limit on the same side of the mean.
- Rule 3: No four out of five consecutive points fall beyond the 0.67σ limit on the same side of the mean.
- Rule 4: No sequence of eight consecutive points appears on one side of the centreline.

These observations indicate that all data points fall within the control limits, and no non-random patterns or trends are evident. This suggests the process is statistically in control.

However, it's important to interpret this conclusion within the correct context. This process focuses solely on inspecting incoming raw materials. Since Tangram Inc.'s manufacturing operations have not yet commenced, labeling the process as "in control" offers limited insight into Tangram Inc.'s own production quality. Instead, the chart reflects supplier performance, not internal process stability. The best this chart can do is suggest something about Tangram Inc.'s manufacturing process, specifically, that the input quality may not pose immediate concerns, assuming the manufacturing process does not introduce additional variation.

The operator's role in this context was purely to inspect components; thus, the chart captures only defect counts, not manufacturing variation. Additionally, without a formal supplier quality agreement in place, it remains uncertain whether the observed defect levels are acceptable. While the np-chart is effective in tracking defect counts, it does not confirm whether the supplier's process is stable or meets required standards. Therefore, this chart should be used cautiously and only for monitoring raw material quality, not for evaluating Tangram Inc.'s internal production processes.

The ideal outcome for this chart would be for all plotted points to be zero or as close to zero as possible, indicating that no defects were found among the five inspected puzzle pieces.

Given these considerations, it is clear that a control chart is not the most suitable tool for analysing Defect Identification Process 1. A more effective tool for this purpose is a Pareto chart,

which can help identify which specific defects occur most frequently. Using the data from Table 1, the Pareto chart below was developed.

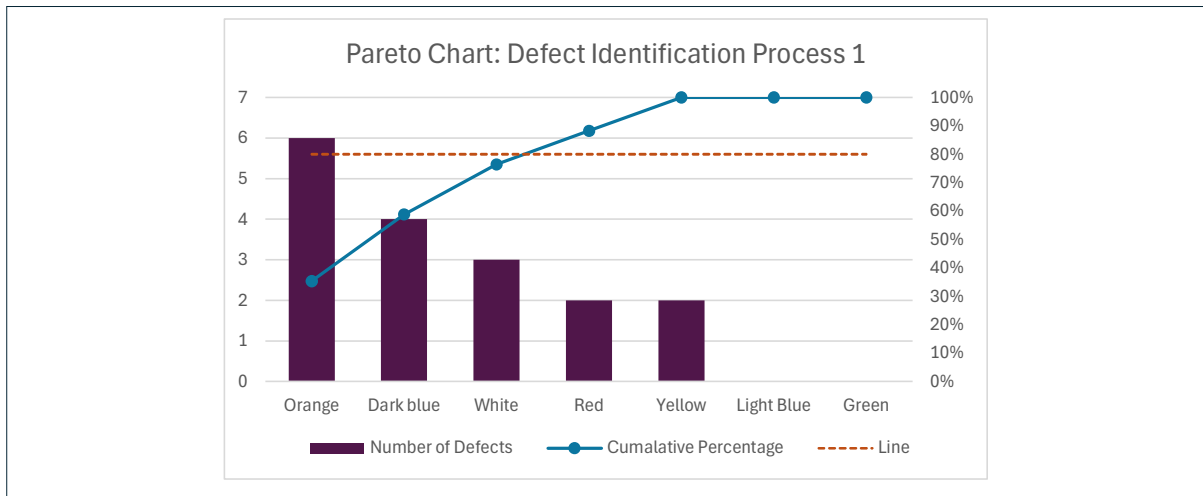


Figure 5: Pareto Chart: Defect Identification Process 1

It is evident from the Pareto chart in Figure 5 that the highest number of defects are Orange, Dark Blue, and White puzzle pieces. These three colours account for approximately 80% of the defects identified during the pre-assembly inspection. This chart can be used in real-time at Tangram Inc.'s manufacturing plant to monitor the quality of incoming raw materials and support effective quality control at the inspection stage.

Based solely on the analysis of Defect Identification Process 1, I would recommend Tangram Inc. source new suppliers for Orange, Dark Blue, and White puzzle pieces, or re-negotiate quality terms with current suppliers to reduce defects

6.1.2 Defect Identification Process 2 Analysis

No formal sampling was conducted in this process. Instead, the operator randomly selected puzzle pieces and placed them on the wooden board. When a defect was identified, the entire semi-assembled Tangram was discarded. A g-chart is appropriate for this scenario, as it is used to monitor the number of units between infrequent nonconforming events, assuming the data follow a geometric distribution (Montgomery, 2019). A geometric distribution models the number of independent Bernoulli trials needed to achieve the first success, where each trial has the same probability of success (Montgomery, 2019). In simpler terms, it tracks how many items are inspected before a defect occurs. Table 2, that allows us to determine the number of puzzle pieces selected before a defect was identified, was used to construct the g-chart below.

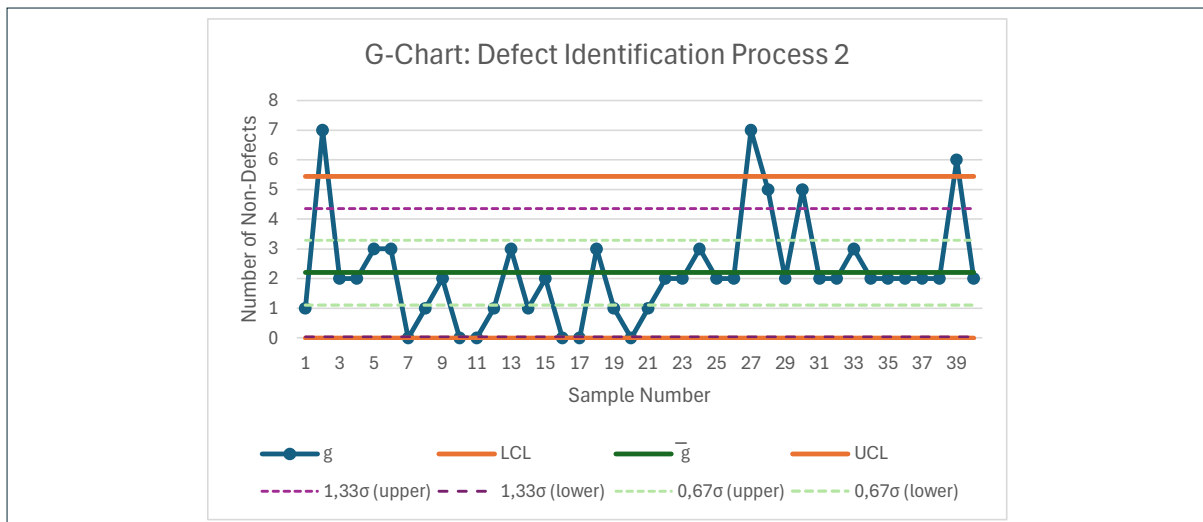


Figure 6: G-Chart: Defect Identification Process 2

The Lower Control Limit (LCL) is 0, indicating that the shortest possible interval between defects is immediate, that is, the very first puzzle piece selected is defective. The Centreline (CL) at 2.2 represents the average number of puzzle pieces selected before encountering a defect, while the Upper Control Limit (UCL) at approximately 5.45 reflects the longest expected interval between defects under normal conditions.

The interpretation of this G-chart, based on the four standard rules, is as follows:

- Rule 1: Three points lie outside the 2σ control limits.
- Rule 2: Two out of three consecutive points (27 and 28, among 26–28) fall beyond the 1.33σ limit on the same side of the mean.
- Rule 3: No group of four out of five consecutive points falls beyond the 0.67σ limit on the same side of the mean.
- Rule 4: There are no runs of eight consecutive points on one side of the centreline.

Based on Rule 1 alone, the process is clearly not in statistical control, indicating the presence of assignable causes of variation. Rule 2 further supports this conclusion, reinforcing that the process exhibits non-random variation.

This chart must be interpreted carefully. Each plotted point represents the number of puzzle pieces selected before identifying a defect. Since a Tangram puzzle consists of 7 pieces, the ideal scenario would be an interval of 7, meaning a complete puzzle with no defects.

G-charts are generally used in processes where defects are rare, so longer intervals between defects are expected and desirable. However, in this case, the opposite is true, finding a defect is common, making longer intervals rare. As a result, points representing defect-free puzzles (an

interval of 7) fall above the UCL, and although statistically "unusual," this is not a good sign. It reflects that defect-free puzzles are rare, which indicates poor raw material quality.

It's also important to state that this chart does not measure Tangram Inc.'s production performance. It strictly monitors the quality of incoming raw materials based on how frequently defects are detected. Shorter intervals (lower values on the G-chart) indicate more frequent defects, whereas longer intervals (higher values) suggest fewer defects. Therefore, while the chart can offer some insight into what Tangram's manufacturing process might encounter, it does not reflect the actual production process itself. The best this chart can do is suggest potential implications for manufacturing, based on supplier quality.

Once again, this highlights that a control chart, while useful in some contexts, is not the most suitable control tool for analysing this defect identification process. A more appropriate tool is the Pareto chart, which identifies which types of defects are most prevalent. Using the data from Table 2, the Pareto chart below was developed.

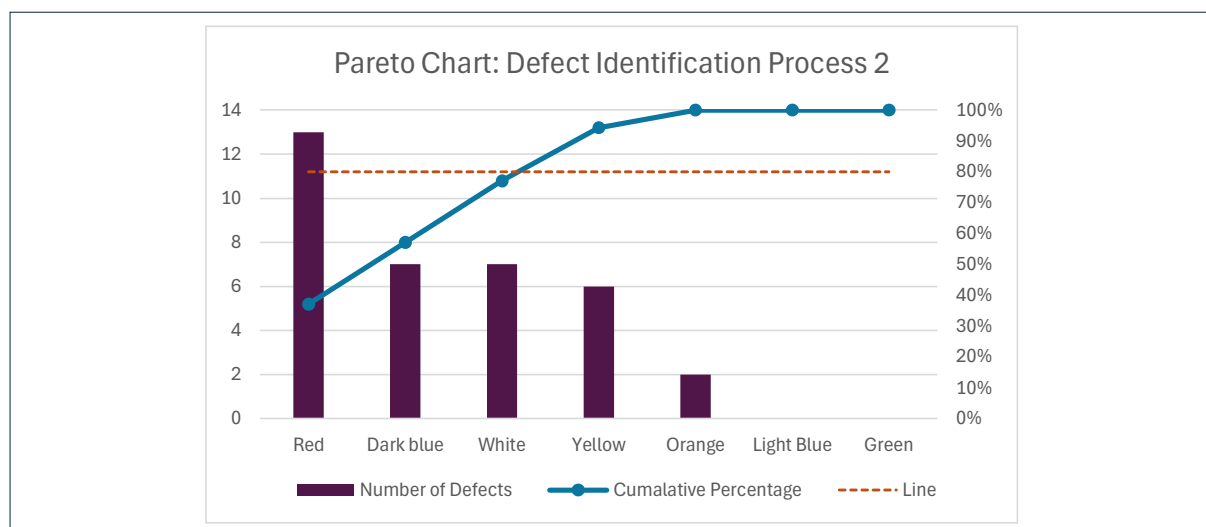


Figure 7: Pareto Chart: Defect Identification Process 2

It is evident from the Pareto chart in Figure 7 that the highest number of defects are Red, Dark Blue, and White puzzle pieces. These three colours account for approximately 80% of the defects identified. This chart can be used in real-time at Tangram Inc.'s manufacturing plant to monitor the quality of raw materials.

Based solely on the analysis alone of Defect Identification Process 2, I would recommend Tangram Inc. source new suppliers for Red, Dark blue and White puzzle pieces or re-negotiate their quality terms to reduce defects.

6.2 Assembly Process Analysis

The main objective of the analysis of these two assembly processes is to determine which produces the best quality output, with defect identification done either before or during assembly. In the context of manufacturing output, quality is often measured by the absence of defects and the consistency of the product meeting the intended standards (Juran & Godfrey, 1999). Thus, a high-quality manufacturing output in our context refers to a completed tangram puzzle that has no puzzle pieces that are defects or in other words tangram sets that aren't defective. To analyse this, we will first look at the process's stability through control chart, and then secondly the capability of each process. Note that cost will be ignored in this preliminary analysis and considered only later.

Figure 3 below illustrates that the analysis will now focus on the entire assembly processes. As the output of manufacturing processes are good indicators of the process the focus will be on the finished goods, the tangram sets.

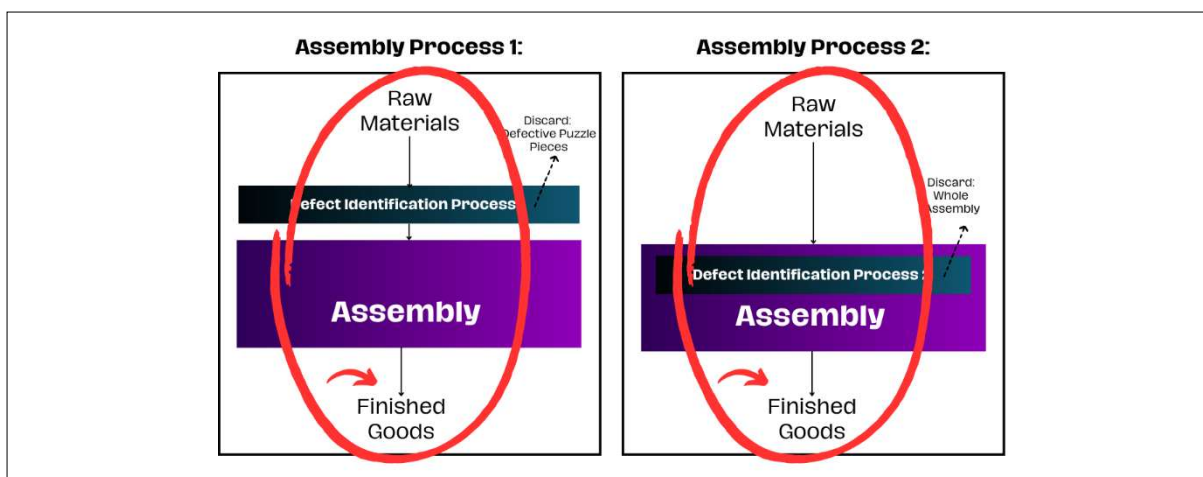


Figure 8: Assembly Processes

6.2.1 Assembly Process 1

6.1.1.1 Assembly Process 1 Stability

Sampling and inspecting before assembly was the strategy used in this assembly process to reduce defects in completed tangram puzzles. Sampling however does not ensure the final product will be free of defects.

Each completed tangram set consistently contained seven puzzle pieces, making the sample size uniform, and the number of defects per tangram set was counted. Thus, a c-chart, which is used to monitor the number of defects per unit in processes where the sample size remains constant (Montgomery, 2013), is suitable for this assembly process.

Using the data from table 1 the following c-chart was constructed.

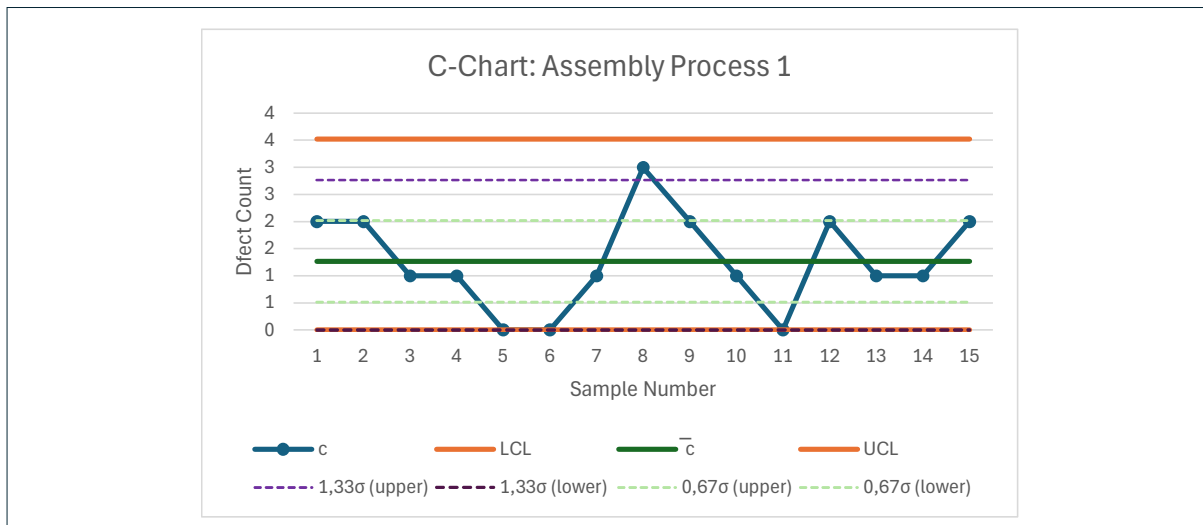


Figure 9: C-Chart: Assembly Process 1

The Lower Control Limit (LCL) is 0, which makes intuitive sense, you cannot have a negative number of defects. The Centreline (CL) is 1.2667, representing the average number of defects per unit, while the Upper Control Limit (UCL) is 3.5175.

The interpretation of this C-chart, based on the four standard control chart rules, is as follows:

- Rule 1: No points lie outside the 2σ control limits.
- Rule 2: No two out of three consecutive points fall beyond the 1.33σ limit on the same side of the mean.
- Rule 3: No group of four out of five consecutive points falls beyond the 0.67σ limit on the same side of the mean.
- Rule 4: There are no runs of eight consecutive points on one side of the centreline.

These observations indicate that all data points fall within the control limits, and no patterns or trends are evident. This suggests the process is statistically in control.

However, it is very important to refer back to the differentiation between defect and defective. A Tangram puzzle consists of seven pieces, so a single nonconforming puzzle piece is counted as one defect, but that one defect renders the entire tangram set defective and unsellable. Therefore, even a small number of defects can have a significant operational impact. This nuance may explain why Tangram Inc. struggles to meet demand despite appearing “in control” statistically.

Given this context, the C-chart may not be the most appropriate tool for assessing the quality of the final product. Instead, a P-chart offers a better fit. A P-chart monitors the proportion of

defective units in a sample, where each unit is classified as either defective or non-defective (Montgomery, 2013).

By organising the data into subgroups of three Tangram sets and categorising each set as either defective or non-defective, we can assess the proportion of defective products in each sample. Using data from Table 1, the P-chart shown below was constructed.

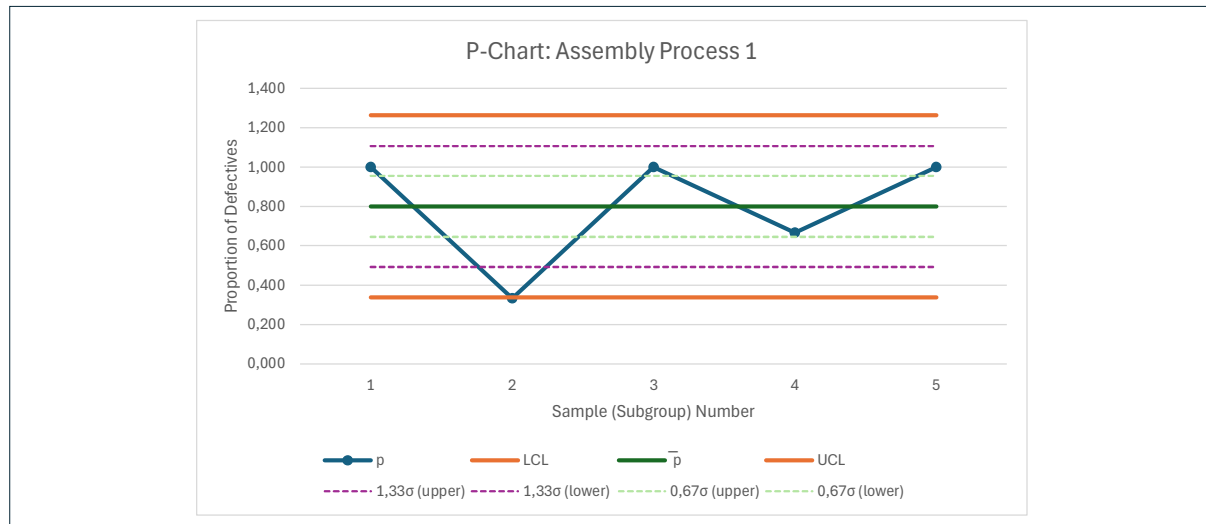


Figure 10: P-Chart: Assembly Process 1

The centreline ($\bar{p}$) represents the average number of defects per unit, while the upper control limit (UCL) and lower control limit (LCL) define the expected range of natural variation in the process. The interpretation of this P-chart, based on the four standard control chart rules, is as follows:

- Rule 1: One point lies outside the 2σ control limits.
- Rule 2: No two out of three consecutive points fall beyond the 1.33σ limit on the same side of the mean.
- Rule 3: No group of four out of five consecutive points falls beyond the 0.67σ limit on the same side of the mean.
- Rule 4: There are no runs of eight consecutive points on one side of the centreline.

Subgroup 2 falls slightly below the lower-control limit, which indicates an unusually low proportion of defectives for that sample. Therefore, the process is classified as being statistically out of control.

However, good practice in statistical quality control recommends collecting data from at least 20 to 25 subgroups to establish stable and accurate control limits (Uyemura, n.d.). With fewer subgroups, the control limits may not accurately reflect the true process variability, which could

lead to unreliable conclusions about process control. This recommendation is also stated in literature addressing the design of statistical control charts in healthcare, which notes that smaller sample sizes can reduce the chart's ability to detect process shifts effectively (Thor et al., 2007).

Therefore, while the absolute best control chart to use to determine the overall process quality, would be a p-chart, the limited number of tangram sets analysed means we cannot confidently determine whether the process is in statistical control. More data is needed to make reliable conclusions.

6.1.1.2 Assembly Process 1 Capability

Although, the importance of defects and defectives was just highlighted above, the quality management team specifically stated that the process should run at a 2-sigma level in terms of defective parts per million. The word **parts** are assumed to refer to puzzle pieces, therefore we will use defects instead of defectives when calculating process capability. Table 1 shows that 15 tangram sets were completed, with a total of 19 defects identified. Each tangram set consists of 7 puzzle pieces, providing 7 opportunities for a defect per set. Based on this, the Defects Per Million Opportunities (DPMO) can be calculated with the following formula.

$$DPMO = \frac{\text{Total number of defects}}{\text{number of units} \times \text{number of opportunities}} \times 1\,000\,000$$

$$DPMO = \frac{19}{(15)(7)} \times 1\,000\,000$$

$$DPMO = 180\,952$$

The calculated DPMO of approximately 180 952 means that, out of every one million opportunities for defects, around 180 952 defects are expected to occur in the tangram assembly process.

Referring to the quality standard set by the quality management team, the process is required to operate at a minimum of a 2-sigma level in terms of defects per million opportunities (DPMO). As stated earlier, this corresponds to a maximum of 308 537 defects per million. Since our calculated DPMO is lower than this threshold, we can conclude that the process meets the minimum quality requirement and is performing at or above the 2-sigma level, but more data points are necessary to confidently conclude such a statement.

6.2.2 Assembly Process 2

6.2.2.1 Assembly Process 2 Stability

A u-chart is used to monitor the number of defects per unit when the sample size varies between observations (Montgomery, 2013). In the tangram assembly process described, no inspection took place before assembling the set, and puzzle pieces were randomly placed on the board. If a defective piece was identified after placement, the entire semi-assembled unit was discarded. Since the number of puzzle pieces placed before discarding varies for each sample, the sample size is not constant, the defects per attempted tangram assembly are counted. Therefore, a u-chart is appropriate here because it adjusts for varying sample sizes by tracking the defect rate (defects per attempted unit) rather than just the count of defects.

Using the data from table 2 the following u-chart was constructed.

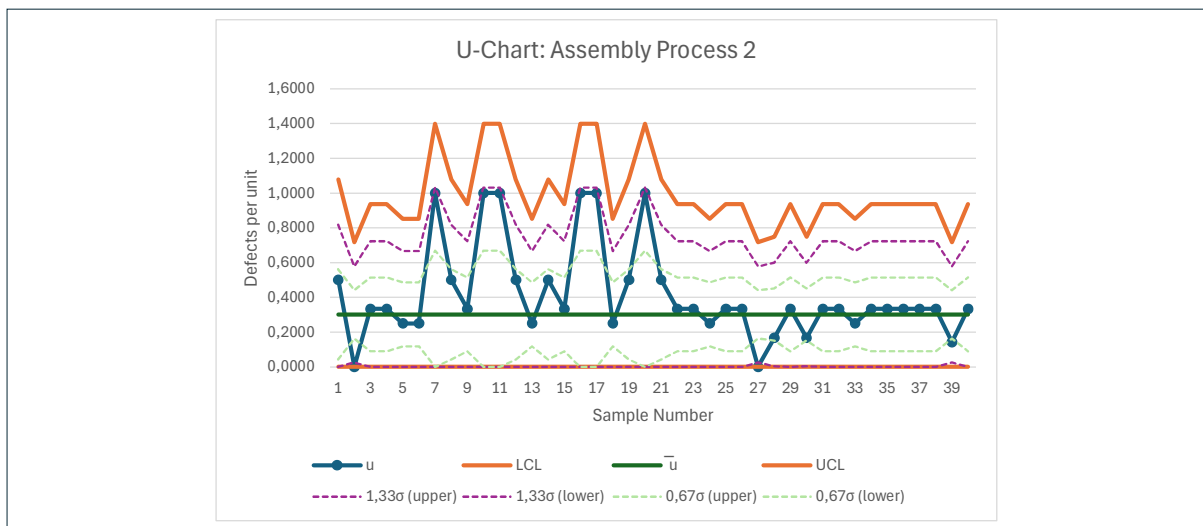


Figure 11: U-Chart: Assembly Process 2

The U-chart for Assembly Process 2 displays the number of defects per unit across 40 samples. The average defect rate ($\bar{u}$) is approximately 0.27, as indicated by the green centreline. The interpretation of this U-chart, based on the four standard control chart rules, is as follows:

- Rule 1: No point lies outside the 2σ control limits.
- Rule 2: No two out of three consecutive points fall beyond the 1.33σ limit on the same side of the mean.
- Rule 3: No group of four out of five consecutive points falls beyond the 0.67σ limit on the same side of the mean.
- Rule 4: There are no runs of eight consecutive points on one side of the centreline.

Based on these observations the process seems to be in statistical control with no points plotted outside of the control limits and no trends or runs identified.

It is important to note that, based on how this process is designed, a defective tangram set should never be completed. However, the current control chart does not account for defectives, only the number of defects identified per attempted Tangram build. To better assess whether the process is truly in control, a P-chart would be more appropriate.

In a P-chart, each completed tangram set would be grouped into subgroups, and the proportion of defective sets within each group would be calculated. Under ideal conditions, there should be no defective sets, since the process is structured to prevent defective products from reaching completion. Therefore, the presence of even a single defective set would point to an assignable cause of variation, such as an operator who may have difficulty detecting defects, perhaps due to poor eyesight or lack of training.

However, in this case, only two Tangram sets were completed during the assembly process. As previously mentioned in the analysis of the earlier process, this sample size is insufficient for constructing a meaningful P-chart. As a result, we must rely on the U-chart for drawing conclusions about process performance at this stage.

6.2.2.2 *Assembly Process 2 Capability*

To calculate the Defects Per Million Opportunities (DPMO) for Assembly Process 2, the total number of defects was divided by the total number of opportunities, where an "opportunity" is defined as a placed puzzle piece. Since the number of puzzle pieces placed per attempted tangram assembly varied, the total number of opportunities was calculated by summing all placed pieces across all samples. This approach aligns with Six Sigma methodology, which states that when the number of opportunities per unit is not consistent, the DPMO formula should be adapted to use total opportunities across all units (Six Sigma Online, n.d.; Lean 6 Sigma Hub, n.d.).

$$DPMO = \frac{\text{Total number of defects}}{\text{Total number of opportunities}} \times 1\,000\,000$$

$$DPMO = \frac{38}{126} \times 1\,000\,000$$

$$DPMO = 301\,587$$

The calculated DPMO of approximately 301 587 means that, out of every one million opportunities for defects, around 301 587 defects are expected to be found in the tangram sets.

Referring to the quality standard set by the quality management team, the process is required to operate at a minimum of a 2-sigma level in terms of defects per million opportunities (DPMO). As stated earlier, this corresponds to a maximum of 308 537 defects per million. Since our calculated DPMO is lower, than this threshold, we can conclude that the process meets the minimum quality requirement and is performing at or above the 2-sigma level. It is very important to note that it is operating very close to the threshold.

6.3 Cost Analysis

The cost analysis will be conducted on both assembly process 1 and 2, focusing on the data collected during the 15-minute experiment for both. Table 3 below lists cost used for this analysis.

Table 4: Cost Information

Item	Cost
Puzzle Piece (Defects)	R2
Wooden Board	R5
Defective semi-assembled item	R20

Tangram Inc. sells a tangram set for R25. From this table above, we can calculate that for each completed tangram set, Tangram Inc. makes a profit of R6 ($R2 \times R7 - R5$).

Assembly Process 1:

For the purpose of this analysis, we will assume that the defective tangram sets are taken apart. The defects are discarded and the wooden board along with the non-defect puzzle pieces are used again.

Table 5: Cost Analysis of Assembly Process 1

15 Samples	Number	Total Cost
Defects Discarded from Sampling	18	(R36)
Defects Discarded from Tangram Set	19	(R19)
Non-defective Tangrams Completed (Profit Only)	3	R18
Loss		(R37)

Assembly Process 2:

Table 6: Cost Analysis of Assembly Process 2

40 Samples	Number	Total Cost
Defective semi-assembled item	38	(R760)
Non-defective Tangrams Completed (Profit Only)	2	R12
Loss		(R748)

Although assembly process 2 resulted in 40 samples within 15 minutes compared to only 15 samples in process 1, this does not necessarily indicate a faster or more efficient process. In process 2, each sample represents an attempted tangram set, many of which consisted of only one or two puzzle pieces before being discarded due to a defect. In contrast, process 1 completed 15 full tangram sets, even though most were ultimately unusable. Therefore, sample counts alone are not meaningful in this context. Instead, the focus should be on the number of usable tangram sets produced and the associated cost or profit over the 15-minute period.

Process 1 produced three non-defective tangram sets at a loss of R37, whereas process 2 produced only two usable sets at a significantly higher loss of R748. A main distinction is that process 1 discards only the defective puzzle piece, minimising waste, while process 2 discards the entire semi-assembled tangram once a defect is detected, regardless of how many non-defective pieces it contains.

Based solely on the cost analysis, assembly process 1 is the more desirable process by far.

7. Conclusion

Overall, the main difference between Assembly Process 1 and Assembly Process 2 lies in the timing of raw material screening and the defect discarding strategy. Assembly Process 1 screens materials *before* manufacturing and discards only individual defect, whereas Assembly Process 2 screens materials *during* manufacturing and discards the entire semi-assembled tangram.

Although Assembly Process 1 requires more data points to confidently interpret control charts, preliminary analysis suggests that the process is currently in statistical control. This conclusion is supported by a C-chart, which tracks the number of defects per tangram set, and a P-chart, which monitors the proportion of defectives per subgroup.

Assembly Process 1 also incorporates Defect Identification Process 1, which appears to be in control based on an NP-chart tracking the number of defectives per inspection. However, further analysis revealed that a Pareto chart provided more valuable insights at this stage. The chart, used to assess supplier quality, showed that approximately 80% of defects were associated with the colours Orange, Dark Blue, or White.

Overall, Assembly Process 1 has a Defects Per Million Opportunities (DPMO) of 180 952, placing it within the 2-sigma performance level. In terms of output, this process produced three non-defective tangram sets in a 15-minute period, with a material loss valued at R37.

In comparison, Assembly Process 2 also appears to be in statistical control, based on analysis using a U-chart, which monitors the number of defects per attempted tangram set.

However, Defect Identification Process 2, used in Assembly Process 2, does not appear to be in control based on a G-chart, which tracks the number of puzzle pieces selected before a defect is found. As with Process 1, a Pareto chart provided deeper insight into supplier-related issues. It revealed that approximately 80% of defects were Red, Dark Blue, or White.

Assembly Process 2 has a DPMO of 301 587, which still falls within the 2-sigma performance level, but is close to the threshold of 308 537; this indicates that assembly process 1 is more capable than assembly process 2. In terms of output, this process produced two non-defective tangram sets in a 15-minute period, with a material loss valued at R748.

The table below summarises the comparison between the two assembly processes.

Table 7: Assembly Process Comparison

	Assembly Process 1	Assembly Process 2
Control Charts	Requires more data, but both assembly process and defect identification process are in control.	Assembly process is in statistical control; however the defect identification process is not.
Process Capability (DPMO) [maximum of 308 537]	180 952	301 587
Defect Identification Strategy	Screening	No screening
Most defects	Orange, Dark blue, White	Red, Dark blue, White

Non-defective Tangrams in 15 minutes	3	2
Loss during the 15 minutes	(R37)	(R748)

The figure below visually represents the conclusion

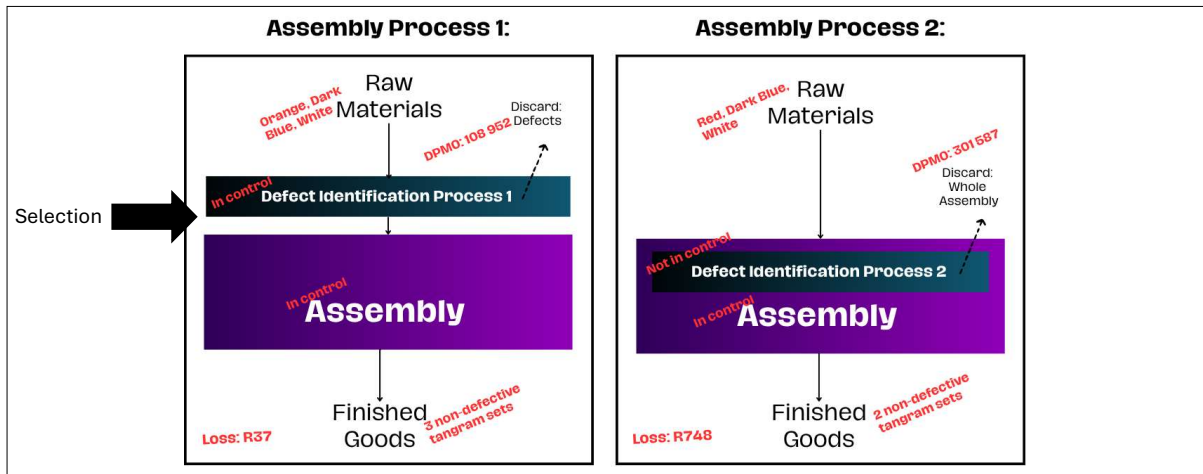


Figure 12: Conclusion summary

Thus, between both assembly processes, I would recommend assembly process 1 along with screening of raw materials, as both the defect identification process and assembly process seem to be in control subject to more data. Assembly process 1 has a lower DPMO than assembly process 2 which means it has a higher process capability. Assembly process 1 was able to complete more non-defective tangram sets than assembly process 2 in 15-minutes, and at a much lower loss. Assembly process 1 thus results in the better outcome considering quality and cost. I would further recommend changing suppliers of Orange, Dark blue, Red and White puzzle pieces or re-negotiating quality terms, to even further improve assembly process 1 and reduce defects and ultimately defectives.

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